IMEN266 Operations Research II
Chapter 8. Queueing Theory

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

1. Queueing Theory
2. Infinite capacity queueing systems
3. Finite capacity queueing systems
4. Variations and non-Markovian queues
5. Time reversibility
6. Queueing network

How to study this chapter (2026 format)

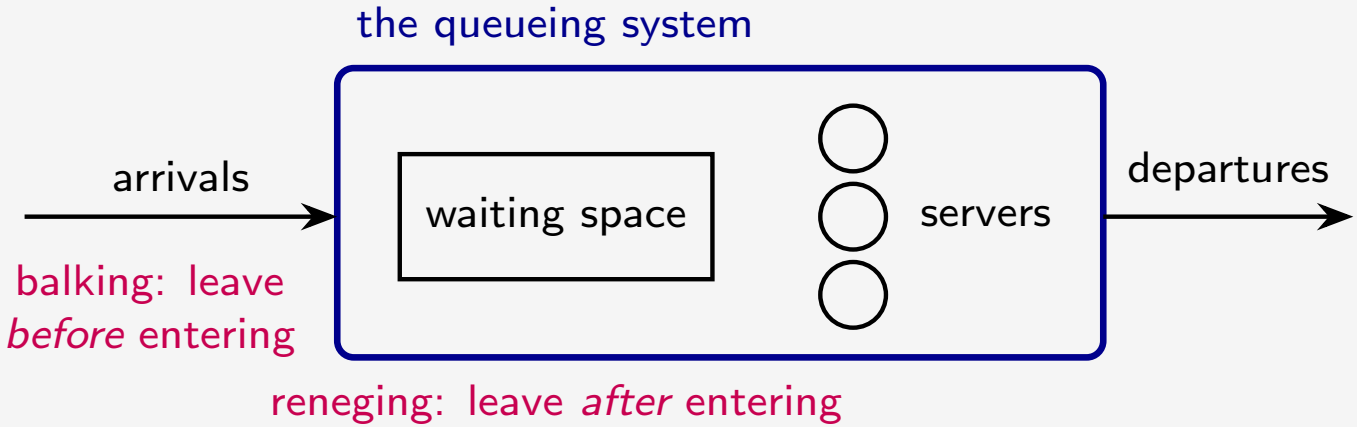
- **Slides** (this file): concepts, the queueing models, and the worked derivations developed in class (p_n, L, W of every model). No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP11 Markovian queues · CP12 finite capacity, $M/G/1$, variations · CP13 Jackson networks): in class the `INCLASS` version (fill the `TODO`s, `check()` answers instantly); at home the full version with solutions.
- **Companion** (gap-fill $\rightarrow$ full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 5**: A exam-style (no AI first) · B critique an AI solution · C stress-test the formulas by discrete-event simulation · D prompt log.
- Class rhythm (75 min): quiz 5′ $\rightarrow$ concepts $\rightarrow$ live notebook $\rightarrow$ wrap-up. Ch. 8 = weeks 12–14.
- Ground truth = analytic $\leftrightarrow$ simulation agreement, not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

History

- 1909 A. K. Erlang (1878-1929), a Danish engineer
  Modeling the dynamics of a telephone switch.
- 1951 David G. Kendall (1918-2007), English statistician
  "Some problems in the theory of queues",
  Journal of the Royal Statistical Society, vol. 31, pp 151-185
  first used the term *Queueing System*.
- 1953 *Kendall notation*
- 1970 Donald L. Iglehart, *Limit theorems for queues*
- 1981 Marcel Neuts, *Matrix Analytic Method*
- 1981 S. Halfin, W. Whitt, *Heavy traffic limit*

Kendall notation

The anatomy of a queueing system



To specify: the **arrival process**; the **service-time** law; the **number of servers**; the **capacity** (waiting space + servers); the **service discipline** (FCFS, LCFS, SPTF = shortest processing time first, SRTF = shortest remaining time first, . . .).

▷ Kendall's shorthand lists exactly these five items: $A/B/s/K/D$.

Kendall notation

The five fields $A/B/s/K/D$

arrivals / service time / # servers / capacity / discipline

A **arrivals**	B **service**	s	K	D
M Poisson process	M exponential	$1, 2, \ldots$	$1, 2, \ldots$	FCFS (default)
G general, GI general independent (= renewal)	G, GI	s	K	LCFS
D deterministic	D	∞	∞ (default)	SPTF
E_k Erlang-k, PH phase type	E_k, PH			SRTF

Two conventions for K

Here K counts the whole *system* (in service + waiting): $M/M/1/K$ holds at most K customers. Some books count the waiting space only.

Kendall notation

Reading examples

$M/M/1$: Poisson arrivals, exponential service, one server, infinite capacity, FCFS (omitted fields take their defaults).
$M/E_3/3/7/LCFS$: Poisson arrivals, Erlang-3 service, 3 servers, at most 7 in the system, last come first served.
$G/G/1+G$: "$+G$" is a non-standard tag for a patience-time law — balking and reneging have no standard notation.

What we want from the analysis

Traditional metrics: the number in the system (queue length) and the waiting time — L, L_q, W, W_q, next slide. Since COVID-19 also the *infection risk* (how many others one overlaps with, for how long), and the psychology of queues: perceived time, information, fairness.

▷ This chapter: FCFS, patient customers; balking/reneging return as CTMCs in CP12 §7 and HW5.

Queueing notation

- λ: average arrival rate
- μ: average service rate
- L: average number of customers in the system
- L_q: average number of customers in the queue
- W: average waiting time of a customer in the system
- W_q: average waiting time of a customer in the queue
- Little's law

$$L = \lambda W$$
$$L_q = \lambda W_q.$$

▷ Companion §1.1 and Proof 3: Little's law $L = \lambda W$ is a renewal reward identity (Ch. 7); $L = L_q + \lambda/\mu$, $W = W_q + 1/\mu$.

Queueing notation

Time averages and customer averages

$X(t) =$ number in the system at time t (waiting + in service), $X_q(t) =$ number waiting; $W_n =$ sojourn time of the n-th customer (wait + service), $W_{q,n} =$ its waiting time:

$$L = \lim_{t\to\infty} E[X(t)], \qquad L_q = \lim_{t\to\infty} E[X_q(t)] \qquad \text{(time averages)},$$

$$W = \lim_{n\to\infty} E[W_n], \qquad W_q = \lim_{n\to\infty} E[W_{q,n}] \qquad \text{(customer averages)}.$$

Three identities

A service lasts $1/\mu$ on average, so $W = W_q + \frac{1}{\mu}$; and $L = L_q + \frac{\lambda}{\mu}$, where $\lambda/\mu =$ mean number *in service* $=$ mean number of busy servers. Little's law $L = \lambda W$, $L_q = \lambda W_q$ ties a time average to a customer average (Companion Proof 3).

▷ Plan for every $M/M/\cdot$ model: $p_n \to L = \sum_n np_n \to$ the other three.

$M/M/1$ queue

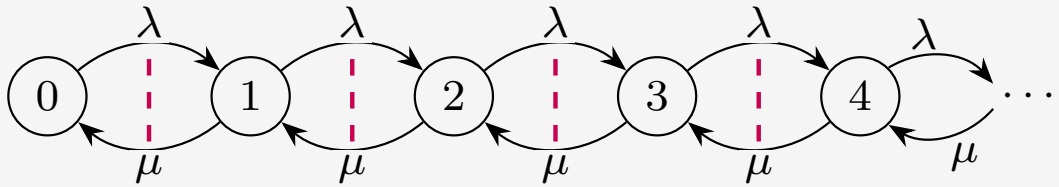
- $X(t) \equiv \#$ of customers in the system at time t.
- $S = \{0, 1, 2, \ldots\}$

▷ Companion Proof 1: birth–death stationary distributions from the cut equations; stability $\rho = \lambda/\mu < 1$.

▶ Colab notebook: CP11 §1 ($M/M/1$, Little, PASTA) `[INCLASS] [full]`

$M/M/1$ queue

Rate diagram and the cut equations



$PP(\lambda)$ in,
one $\exp(\mu)$ server

A birth–death CTMC ($\lambda_n = \lambda$, $\mu_n = \mu$). Across each dashed cut the two flows balance:

$$\lambda p_0 = \mu p_1 \;\Rightarrow\; p_1 = \tfrac{\lambda}{\mu}\, p_0, \qquad \lambda p_1 = \mu p_2 \;\Rightarrow\; p_2 = \tfrac{\lambda}{\mu}\, p_1 = \left(\tfrac{\lambda}{\mu}\right)^2 p_0,$$

$$\lambda p_2 = \mu p_3 \;\Rightarrow\; p_3 = \left(\tfrac{\lambda}{\mu}\right)^3 p_0, \qquad \dots \qquad p_n = \left(\tfrac{\lambda}{\mu}\right)^n p_0, \quad n \geq 0$$

(p_0 from normalisation, next slide).

$M/M/1$ queue

Normalisation and the stability condition

$$p_0 + p_1 + p_2 + \cdots = p_0\left(1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \cdots\right) = 1.$$

The geometric series converges iff $\rho := \lambda/\mu < 1$ (**stability condition**: the server is faster than the arrivals). Then

$$p_0 \cdot \frac{1}{1-\lambda/\mu} = 1 \quad \Longrightarrow \quad p_0 = 1 - \frac{\lambda}{\mu} = 1 - \rho\,, \qquad p_n = (1-\rho)\,\rho^n.$$

Reading ρ

$\rho = \lambda/\mu$ is the **traffic intensity**; $1 - p_0 = \rho$ is the long-run fraction of time the server is busy. For $\rho \geq 1$ no stationary distribution exists — the queue grows without bound (CP11 §1).

$M/M/1$ queue

L from $p_i = (1-\rho)\rho^i$: the derivative trick

$$L = \sum_{i=0}^{\infty} i\, p_i = 0 \cdot p_0 + 1 \cdot p_1 + 2 \cdot p_2 + \cdots = (1-\rho)\,\rho\,\left(1 + 2\rho + 3\rho^2 + \cdots\right).$$

For $0 < x < 1$:

$$1 + 2x + 3x^2 + \cdots = \frac{d}{dx}\left(1 + x + x^2 + \cdots\right) = \frac{d}{dx}\,\frac{1}{1-x} = \frac{1}{(1-x)^2}, \text{ so}$$

$$L = \frac{(1-\rho)\,\rho}{(1-\rho)^2} = \frac{\rho}{1-\rho} = \boxed{\frac{\lambda}{\mu - \lambda}}.$$

▷ The same trick, $\sum_i i\, x^{i-1} = 1/(1-x)^2$, handles every geometric tail below ($M/M/2$, $M/M/s$).

$M/M/1$ queue

W, W_q, L_q by Little's law and the two identities

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}, \qquad W_q = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)},$$

$$L_q = \lambda W_q = \frac{\lambda^2}{\mu(\mu - \lambda)} \qquad \left(\text{check: } L - L_q = \frac{\lambda}{\mu} = \rho\right).$$

Not only the mean: in steady state the sojourn time of a customer has the $\exp(\mu - \lambda)$ distribution (Companion Proof 4: a geometric number of exponential phases).

The blow-up as $\rho \uparrow 1$

$L = \rho/(1-\rho)$: $\rho = 0.5,\ 0.9,\ 0.99 \ \Rightarrow\ L = 1,\ 9,\ 99$. Near $\rho = 1$ a small increase in load multiplies the queue — and a simulation needs a run length growing like $1/(1-\rho)^2$ to measure it (CP11 §1).

$M/M/2$ queue

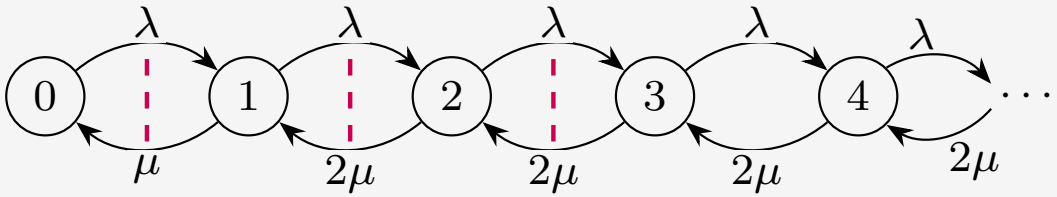
- $X(t) \equiv \#$ of customers in the system at time t.
- $S = \{0, 1, 2, \ldots\}$

▷ Companion Proof 2: $M/M/s$ with $s = 2$; Proof 2(c) compares two servers with one server of double speed.

▶ Colab notebook: CP11 §2 (two servers vs. one fast) `[INCLASS] [full]`

$M/M/2$ queue

Rate diagram: the death rate saturates at 2μ



one busy server $\to \mu$,
both busy $\to 2\mu$

$$\lambda p_0 = \mu p_1 \;\Rightarrow\; p_1 = \tfrac{\lambda}{\mu}\, p_0, \qquad \lambda p_1 = 2\mu p_2 \;\Rightarrow\; p_2 = \tfrac{\lambda}{2\mu}\, p_1 = \tfrac{\lambda}{2\mu} \cdot \tfrac{\lambda}{\mu}\, p_0,$$

$$\lambda p_2 = 2\mu p_3 \;\Rightarrow\; p_3 = \tfrac{\lambda}{2\mu}\, p_2 = \left(\tfrac{\lambda}{2\mu}\right)^2 \tfrac{\lambda}{\mu}\, p_0, \quad \dots$$

$$p_i = \left(\tfrac{\lambda}{2\mu}\right)^{i-1} \tfrac{\lambda}{\mu}\, p_0 \qquad (i \geq 1)$$

▷ Only the first ratio is λ/μ; from state 1 on the ratio is $\lambda/(2\mu)$ — this is what stability will depend on.

$M/M/2$ queue

Normalisation and stability

$$1 = p_0\Big(1 + \tfrac{\lambda}{\mu} + \tfrac{\lambda}{\mu}\tfrac{\lambda}{2\mu} + \tfrac{\lambda}{\mu}\big(\tfrac{\lambda}{2\mu}\big)^2 + \cdots\Big)$$
$$= p_0\Big(1 + \tfrac{\lambda}{\mu}\big(1 + \tfrac{\lambda}{2\mu} + (\tfrac{\lambda}{2\mu})^2 + \cdots\big)\Big).$$

If $\lambda/(2\mu) < 1$ (stability: two servers out-pace the arrivals), the inner series is $1/(1-\lambda/(2\mu))$ and

$$p_0\Big(1 + \frac{\lambda}{\mu}\cdot\frac{1}{1-\lambda/(2\mu)}\Big) = 1 \quad\Longrightarrow\quad p_0 = \frac{2\mu-\lambda}{2\mu+\lambda}.$$

Stationary distribution

$$p_i = \frac{\lambda}{\mu}\Big(\frac{\lambda}{2\mu}\Big)^{i-1}\frac{2\mu-\lambda}{2\mu+\lambda}, \qquad i = 1, 2, 3, \ldots$$

$M/M/2$ queue

L, and the rest as an exercise

$$L = \sum_{i\geq 0} i\, p_i = \frac{\lambda}{\mu}\cdot\frac{2\mu-\lambda}{2\mu+\lambda}\left(1+2\tfrac{\lambda}{2\mu}+3\left(\tfrac{\lambda}{2\mu}\right)^2+\cdots\right)$$

$$= \frac{\lambda}{\mu}\cdot\frac{2\mu-\lambda}{2\mu+\lambda}\cdot\frac{1}{(1-\lambda/(2\mu))^2} = \frac{\lambda}{\mu}\cdot\frac{2\mu-\lambda}{2\mu+\lambda}\cdot\frac{4\mu^2}{(2\mu-\lambda)^2}$$

$$= \frac{4\lambda\mu}{4\mu^2-\lambda^2}.$$

W, W_q, L_q (exercise): $W = \frac{L}{\lambda} = \frac{4\mu}{4\mu^2-\lambda^2}$, $L_q = L - \frac{\lambda}{\mu} = \frac{\lambda^3}{\mu(4\mu^2-\lambda^2)}$, $W_q = \frac{L_q}{\lambda} = \frac{\lambda^2}{\mu(4\mu^2-\lambda^2)}$.

▷ CP11 §2 checks these by simulation and compares two servers of rate μ with one server of rate 2μ.

$M/M/s$ queue

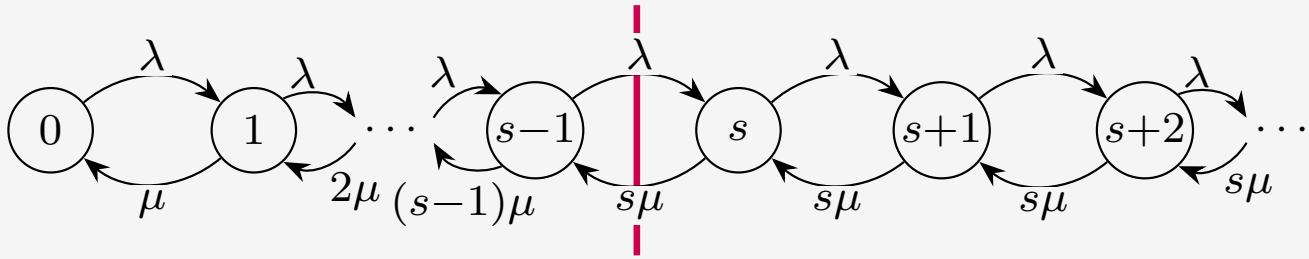
- $X(t) \equiv \#$ of customers in the system at time t.
- $S = \{0, 1, 2, \ldots\}$

▷ Companion Proof 2: p_0, L_q and the Erlang C probability of waiting $P(W_q > 0) = p_s/(1-\rho)$.

▶ Colab notebook: CP11 §3 ($M/M/s$, Erlang C) `[INCLASS] [full]`

$M/M/s$ queue

Rate diagram: death rate $\min(n, s)\,\mu$ — two regimes



Below s (left of the bar; the ratios $\lambda/(n\mu)$ shrink):

$$p_1 = \tfrac{\lambda}{\mu}p_0, \quad p_2 = \tfrac{\lambda}{2\mu}p_1 = \tfrac{1}{2!}\left(\tfrac{\lambda}{\mu}\right)^2 p_0, \quad p_3 = \tfrac{1}{3\cdot 2}\left(\tfrac{\lambda}{\mu}\right)^3 p_0, \quad \dots \qquad p_s = \tfrac{1}{s!}\left(\tfrac{\lambda}{\mu}\right)^s p_0$$

From s *on* (all servers busy, constant ratio $\lambda/(s\mu)$):

$$p_{s+1} = \tfrac{\lambda}{s\mu}p_s, \quad p_{s+2} = \tfrac{\lambda}{s\mu}p_{s+1} = \left(\tfrac{\lambda}{s\mu}\right)^2 p_s, \quad \dots \qquad p_{s+i} = \left(\tfrac{\lambda}{s\mu}\right)^i p_s$$

$M/M/s$ queue

Normalisation: a finite sum plus a geometric tail

$$1 = \sum_i p_i = \underbrace{p_0 + \tfrac{\lambda}{\mu}p_0 + \cdots + \tfrac{1}{s!}\left(\tfrac{\lambda}{\mu}\right)^s p_0}_{\text{states } 0,\ldots,s} + \underbrace{\tfrac{\lambda}{s\mu}p_s + \left(\tfrac{\lambda}{s\mu}\right)^2 p_s + \cdots}_{\text{states } s+1,s+2,\ldots}$$

$$= p_0\Big(\sum_{i=0}^{s} \tfrac{1}{i!}\left(\tfrac{\lambda}{\mu}\right)^i + \tfrac{1}{s!}\left(\tfrac{\lambda}{\mu}\right)^s\big[\tfrac{\lambda}{s\mu} + \left(\tfrac{\lambda}{s\mu}\right)^2 + \cdots\big]\Big).$$

If $\lambda/(s\mu) < 1$ (stability: s servers out-pace the arrivals) the bracket is $\frac{\lambda/(s\mu)}{1-\lambda/(s\mu)}$ and

$$\underbrace{p_0\Big(\sum_{i=0}^{s} \frac{1}{i!}\Big(\frac{\lambda}{\mu}\Big)^i + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s \frac{\lambda/(s\mu)}{1-\lambda/(s\mu)}\Big)}_{=:A} = 1 \implies \boxed{p_0 = \frac{1}{A}}.$$

$M/M/s$ queue

L_q directly: only the states above s have a queue

$$L_q = 0 \cdot p_0 + \cdots + 0 \cdot p_s + 1 \cdot p_{s+1} + 2 \cdot p_{s+2} + \cdots = \sum_{i=1}^{\infty} i\, p_{s+i}$$

$$= \sum_{i=1}^{\infty} i\, p_0 \tfrac{1}{s!} \left(\tfrac{\lambda}{\mu}\right)^s \left(\tfrac{\lambda}{s\mu}\right)^i = p_0 \tfrac{1}{s!} \left(\tfrac{\lambda}{\mu}\right)^s \tfrac{\lambda}{s\mu} \sum_{i\geq 1} i \left(\tfrac{\lambda}{s\mu}\right)^{i-1}$$

$$= \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^s \frac{\lambda}{s\mu}\left(\frac{s\mu}{s\mu-\lambda}\right)^2 p_0$$

(the last sum is $1/(1-\lambda/(s\mu))^2$ by the derivative trick).

Then L, W, W_q — and the probability of waiting

$L = L_q + \lambda/\mu, \quad W_q = L_q/\lambda, \quad W = W_q + 1/\mu;$

$P(\text{wait}) = \sum_{i\geq s} p_i = \dfrac{p_s}{1-\lambda/(s\mu)}$ (**Erlang C**; CP11 §3).

$M/M/\infty$ queue

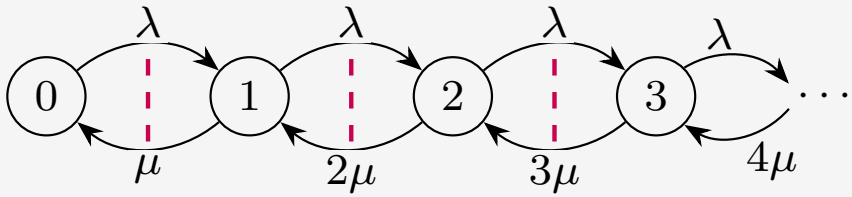
- $X(t) \equiv \#$ of customers in the system at time t.
- $S = \{0, 1, 2, \ldots\}$

▷ Companion Proof 1(c): the Poisson stationary law; Proof 6(c) truncates it to $M/M/K/K$.

▶ Colab notebook: CP11 §4 ($M/M/\infty$) `[INCLASS] [full]`

$M/M/\infty$ queue

Rate diagram: every customer has its own server, death rate $n\mu$



self-service: nobody waits;
n customers $= n$ running clocks

$$p_1 = \tfrac{\lambda}{\mu}\,p_0, \quad p_2 = \tfrac{\lambda}{2\mu}\,p_1 = \tfrac{1}{2!}\big(\tfrac{\lambda}{\mu}\big)^2 p_0, \quad p_3 = \tfrac{\lambda}{3\mu}\,p_2 = \tfrac{1}{3!}\big(\tfrac{\lambda}{\mu}\big)^3 p_0, \quad \dots$$

$$p_i = \frac{1}{i!}\Big(\frac{\lambda}{\mu}\Big)^i p_0, \qquad i = 0, 1, 2, \dots$$

— the first regime of $M/M/s$, which here never ends: no stability condition to check.

$M/M/\infty$ queue

Normalisation gives a Poisson distribution

$$\sum_i p_i = p_0\left[1 + \frac{\lambda}{\mu} + \frac{1}{2!}\left(\frac{\lambda}{\mu}\right)^2 + \frac{1}{3!}\left(\frac{\lambda}{\mu}\right)^3 + \cdots\right] = 1,$$

and $\sum_{i\geq 0}(\lambda/\mu)^i/i! = e^{\lambda/\mu}$, so

$$p_0 = e^{-\lambda/\mu}, \qquad p_i = \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i e^{-\lambda/\mu} = \frac{e^{-\lambda/\mu}(\lambda/\mu)^i}{i!}$$

— the Poisson pmf with mean λ/μ.

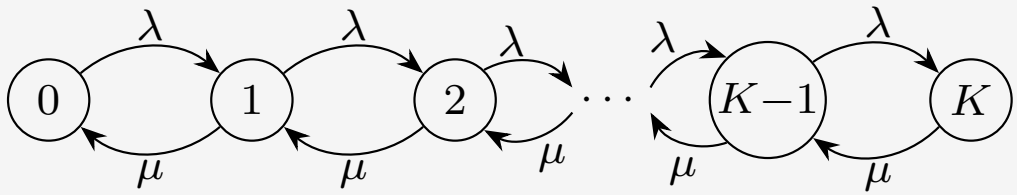
Performance measures

$$L = \sum_i i p_i = \frac{\lambda}{\mu}, \qquad W = \frac{L}{\lambda} = \frac{1}{\mu} \text{ (the service time)}, \qquad L_q = W_q = 0.$$

▷ The Poisson law survives for any service-time distribution with mean $1/\mu$ ($M/G/\infty$) — an insensitivity met again in $M/G/K/K$.

$M/M/1/K$ queue

Finite capacity K: the chain stops at K



$S = \{0, 1, \ldots, K\}$; an arrival that finds K customers is **lost**

The same cut equations as $M/M/1$, but only K of them — a finite sum, so *no stability condition*:

$$p_1 = \tfrac{\lambda}{\mu} p_0, \quad p_2 = \left(\tfrac{\lambda}{\mu}\right)^2 p_0, \quad \ldots, \quad p_K = \left(\tfrac{\lambda}{\mu}\right)^K p_0; \qquad p_0 \sum_{i=0}^{K} \left(\frac{\lambda}{\mu}\right)^i = 1$$

$$\Rightarrow \quad p_0 = \frac{1 - \lambda/\mu}{1 - (\lambda/\mu)^{K+1}}, \qquad p_i = \frac{(1 - \lambda/\mu)\,(\lambda/\mu)^i}{1 - (\lambda/\mu)^{K+1}}, \quad i = 0, \ldots, K$$

($\lambda = \mu$: $p_i = 1/(K+1)$).

- Colab notebook: CP12 §1 ($M/M/1/K$) `[INCLASS] [full]`

$M/M/1/K$ queue

The number that matters, and Little's law with the *effective* arrival rate

$$p_K = \frac{(1-\lambda/\mu)(\lambda/\mu)^K}{1-(\lambda/\mu)^{K+1}}$$ = probability that the system is full = (PASTA) fraction of arrivals that are **dropped**.

$L = \sum_{i=0}^{K} i\, p_i$ (finite sum). $W = L/\lambda$? **No**: a fraction p_K of the arrivals never enters and has no sojourn time. The customers who enter arrive at the **actual (effective) rate** $\lambda(1-p_K)$, so

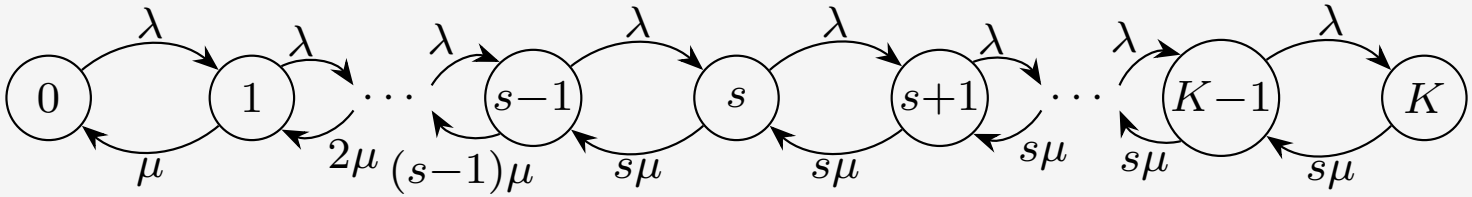
$$W = \frac{L}{\lambda(1-p_K)}, \qquad W_q = W - \frac{1}{\mu}, \qquad L_q = \lambda(1-p_K)\, W_q.$$

Example (CP12 §1)

$\lambda = \mu$, $K = 4$: $p_i = \frac{1}{5}$, $p_K = 0.2$, $L = 2$, $\lambda(1-p_K) = 0.8\lambda$, $W = 2.5/\lambda$ — the naive $L/\lambda = 2/\lambda$ is 20 % too short.

$M/M/s/K$ queue

$s \le K$: the $M/M/s$ diagram cut off at K



The cut equations are those of $M/M/s$ (slides 17–19) up to K — a finite sum, no stability condition:

$$p_i = \begin{cases} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i p_0, & 0 \le i \le s, \\ \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda}{s\mu}\right)^{i-s} p_0, & s \le i \le K, \end{cases} \qquad p_0 = \left(\sum_{i=0}^{K} \frac{p_i}{p_0}\right)^{-1}.$$

- Colab notebook: CP12 §2 ($M/M/s/K$, Example $M/M/5/5$) `[INCLASS] [full]`

$M/M/s/K$ queue

Performance measures — again with the effective arrival rate

$$L = \sum_{i=0}^{K} i\, p_i, \qquad W = \frac{L}{\lambda(1-p_K)}, \qquad W_q = W - \frac{1}{\mu}, \qquad L_q = \lambda(1-p_K)\, W_q$$

($L_q = \sum_{i>s}(i-s)\, p_i$ directly). p_K = fraction of customers lost; $\lambda(1-p_K)$ = throughput = rate of service completions.

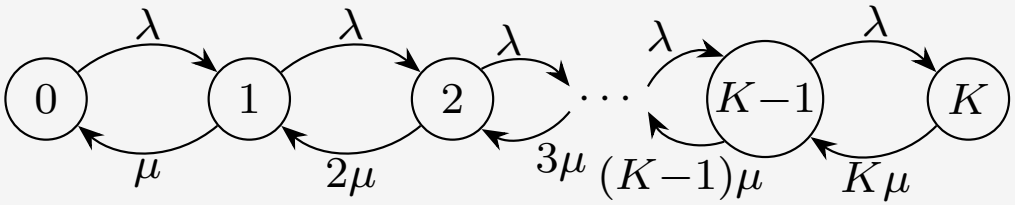
Check

$L - L_q = \lambda(1-p_K)/\mu$ = mean number of busy servers: every served customer occupies a server for $1/\mu$ on average (Little's law for the servers).

▷ Recipe for *every* finite-capacity queue: p_K first, then $\lambda(1-p_K)$, then Little.

$M/M/K/K$ queue

The loss system of A. K. Erlang: K servers, no waiting room



$M/M/\infty$ cut off at K: a call that finds all K lines busy is lost

$$p_1 = \tfrac{\lambda}{\mu} p_0, \quad p_2 = \tfrac{1}{2!}\left(\tfrac{\lambda}{\mu}\right)^2 p_0, \ \ldots, \ p_i = \tfrac{1}{i!}\left(\tfrac{\lambda}{\mu}\right)^i p_0, \ \ldots, \ p_K = \tfrac{1}{K!}\left(\tfrac{\lambda}{\mu}\right)^K p_0;$$

$$p_0 \sum_{i=0}^{K} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i = 1 \quad \Longrightarrow \quad \boxed{p_0 = \left(\sum_{i=0}^{K} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i\right)^{-1}}.$$

- Colab notebook: CP12 §3 (Erlang loss) `[INCLASS] [full]`

$M/M/K/K$ queue

Stationary distribution and the Erlang loss formula

$$p_i = \frac{\frac{1}{i!}(\lambda/\mu)^i}{\sum_{j=0}^{K} \frac{1}{j!}(\lambda/\mu)^j}, \quad i = 0, \ldots, K \qquad \text{— } \textit{works for } M/G/K/K \textit{ too.}$$

$$p_K = \frac{\frac{1}{K!}(\lambda/\mu)^K}{\sum_{j=0}^{K} \frac{1}{j!}(\lambda/\mu)^j}$$

Erlang loss formula (Erlang B)

$=$ fraction of calls that find all K lines busy and are lost; only the mean call length $1/\mu$ enters.

Performance

No waiting: $W = 1/\mu$, $W_q = L_q = 0$, $L = \lambda(1 - p_K)/\mu =$ mean number of busy lines. The insensitivity (same p_i for any service law with mean $1/\mu$) is checked by simulation in CP12 §3.

Example $M/M/5/5$ queue

- $\lambda = 5$ per hour
- $1/\mu = 12$ minutes

1. What fraction of customers is rejected?
2. $L =$?
3. $W =$?

▷ Companion Proof 5: finite capacity — p_K is the loss probability; Little's law needs the effective arrival rate $\lambda(1 - p_K)$.

▶ Colab notebook: CP12 §2 (Example $M/M/5/5$) `[INCLASS] [full]`

Example $M/M/5/5$ queue

Solution: $\lambda = 5$/h, $1/\mu = 12$ min $\Rightarrow \mu = 5$/h, so $\lambda/\mu = 1$

1. Fraction rejected $= p_5$ (Erlang B with $K = 5$, $\lambda/\mu = 1$):

$$p_5 = \frac{\frac{1}{5!}\,1^5}{\sum_{i=0}^{5} \frac{1}{i!}\,1^i} = \frac{1/120}{1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120}} = \frac{1/120}{326/120} = \frac{1}{326}$$

$\approx 0.31\,\%$ of the customers.

2. $L = 0 \cdot p_0 + 1 \cdot p_1 + \cdots + 5 \cdot p_5 = \dfrac{325}{326} \approx 0.997$ (shortcut: $L = \lambda(1 - p_K)/\mu = 1 - \frac{1}{326}$).

3. $W = \dfrac{L}{\lambda(1 - p_K)} = \dfrac{325/326}{5\,(1 - \frac{1}{326})} = \dfrac{325/326}{5 \cdot 325/326} = \dfrac{1}{5}$ h $= 12$ min — nobody waits in a loss system, so $W = 1/\mu$.

▷ Dividing L by λ instead of $\lambda(1 - p_K)$ gives 11.96 min, a sojourn shorter than a service — the sign of a forgotten p_K.

Bulk arrival: airport

- $M^{[X]}/M/1$ queue
- Poisson arrivals: $PP(\lambda)$
- Each arrival brings X customers.

$$P(X = i) = q_i, \text{ for } i = 1, 2, 3, \ldots.$$

- Service is individual.
- Results

$$L = \frac{\rho + \left(\frac{\lambda}{\mu} E\left[X^2\right]\right)}{2(1-\rho)}, \text{ where } \rho = \frac{\lambda}{\mu} E[X].$$

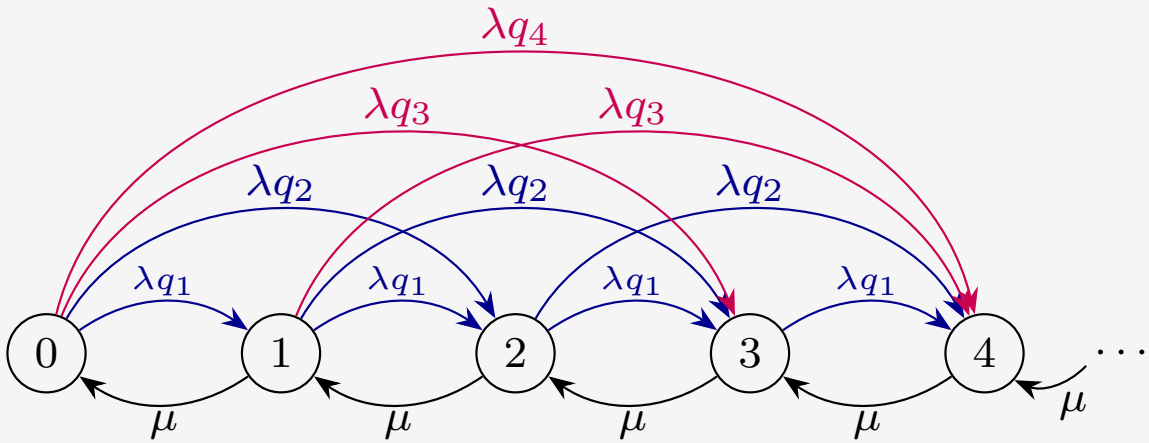
▷ Companion Proof 7(c): the $M^{[X]}/M/1$ formula; $X \equiv 1$ gives back $M/M/1$.

Numbers of the airport example: derivation frame.

▶ Colab notebook: CP12 §6 (bulk arrivals) `[INCLASS] [full]`

Bulk arrival: airport

Rate diagram of $M^{[X]}/M/1$ — not a birth–death chain



A batch of size i (probability q_i) moves the state up by i: arcs from n to $n+1, n+2, \ldots$ with rates $\lambda q_1, \lambda q_2, \ldots$; departures are single, rate μ. A cut now crosses several arcs, so the two-term recursion is gone — the balance equations are solved with a generating function (Companion Proof 7(c)); the slide quotes the result. $X \equiv 1$ gives back $M/M/1$.

▷ Stability: $\rho = \lambda E[X]/\mu < 1$ — customers, not batches, must be served faster than they arrive.

Bulk arrival: airport

Numbers: batches of $1, 2, 3, 4$ w.p. $0.4, 0.3, 0.2, 0.1$; $\lambda = 2$ batches/min, $\mu = 5$/min

$E[X] = 0.4 + 0.6 + 0.6 + 0.4 = 2$, $E[X^2] = 0.4 + 1.2 + 1.8 + 1.6 = 5$, $\rho = \lambda E[X]/\mu = 2 \cdot 2/5 = 0.8$;

$$L = \frac{\rho + \frac{\lambda}{\mu} E[X^2]}{2(1-\rho)} = \frac{0.8 + \frac{2}{5} \cdot 5}{2 \cdot 0.2} = 7 \text{ customers}, \qquad L_q = L - \rho = 6.2,$$

$$W = \frac{L}{\lambda E[X]} = \frac{7}{4} = 1.75 \text{ min}, \qquad W_q = \frac{L_q}{\lambda E[X]} = \frac{6.2}{4} = 1.55 \text{ min}.$$

Two things to notice

Little's law uses the *customer* arrival rate $\lambda E[X] = 4$/min, not the batch rate. Single arrivals at the same customer rate ($M/M/1$, $\rho = 0.8$) would give $L = 4$: batching almost doubles the congestion at the same utilisation (CP12 §6).

$M/G/1$ queue

- Poisson arrivals: $PP(\lambda)$.
- General service times following $G(\cdot)$ with mean $1/\mu$ and variance σ^2.
- Results

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)}.$$

▷ Companion Proof 7: Pollaczek–Khinchine from the mean residual service time (Ch. 7 inspection paradox) + PASTA + Little.

▶ Colab notebook: CP12 §5 ($M/G/1$, P–K formula) `[INCLASS] [full]`

▷ $\rho = \lambda/\mu$. The result comes from the embedded DTMC at departure epochs (the number left behind by the n-th departure is Markov); $G/M/1$ is treated the same way at arrival epochs.

$M/G/1$ queue

Where the formula comes from: a mean-value argument

An arrival waits for (i) the *residual* service time R of the customer in service, if any, and (ii) the full services of the L_q customers ahead of it:

$$W_q = E[R] + L_q \cdot \frac{1}{\mu}.$$

By PASTA the arrival sees time averages; by Ch. 7 (inspection paradox) the residual service averages $E[S^2]/(2E[S])$ *while the server is busy*, which is a fraction ρ of the time:

$$E[R] = \rho \, \frac{E[S^2]}{2E[S]} = \frac{\lambda E[S^2]}{2}, \qquad E[S^2] = \sigma^2 + \frac{1}{\mu^2}.$$

With Little, $L_q = \lambda W_q$: $W_q = \dfrac{\lambda E[S^2]}{2} + \rho \, W_q$, so

$$W_q = \frac{\lambda E[S^2]}{2(1-\rho)}, \qquad L_q = \lambda W_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}.$$

$M/G/1$ queue

Reading the Pollaczek–Khinchine formula

Only the mean and the **variance** of the service time matter: $\sigma^2 = 1/\mu^2$ (exponential) gives back $M/M/1$; $\sigma^2 = 0$ ($M/D/1$) halves L_q; a long-tailed service law can make L_q huge at the same ρ.

Numbers: a web server

$\lambda = 20$ requests/min $= \frac{1}{3}$/s, $E[S] = 2$ s, $\sigma = 5$ s (Pareto-like file sizes): $\rho = \frac{2}{3}$,

$$L_q = \frac{(\frac{1}{3})^2 \cdot 25 + (\frac{2}{3})^2}{2 \cdot \frac{1}{3}} = 4.83, \qquad W_q = 14.5 \text{ s}, \quad W = 16.5 \text{ s}, \quad L = 5.5;$$

with exponential service of the same mean, W_q would be 4 s.

▷ CP12 §5 simulates $M/D/1$, $M/M/1$ and a heavy-tailed $M/G/1$ at the same ρ against the formula.

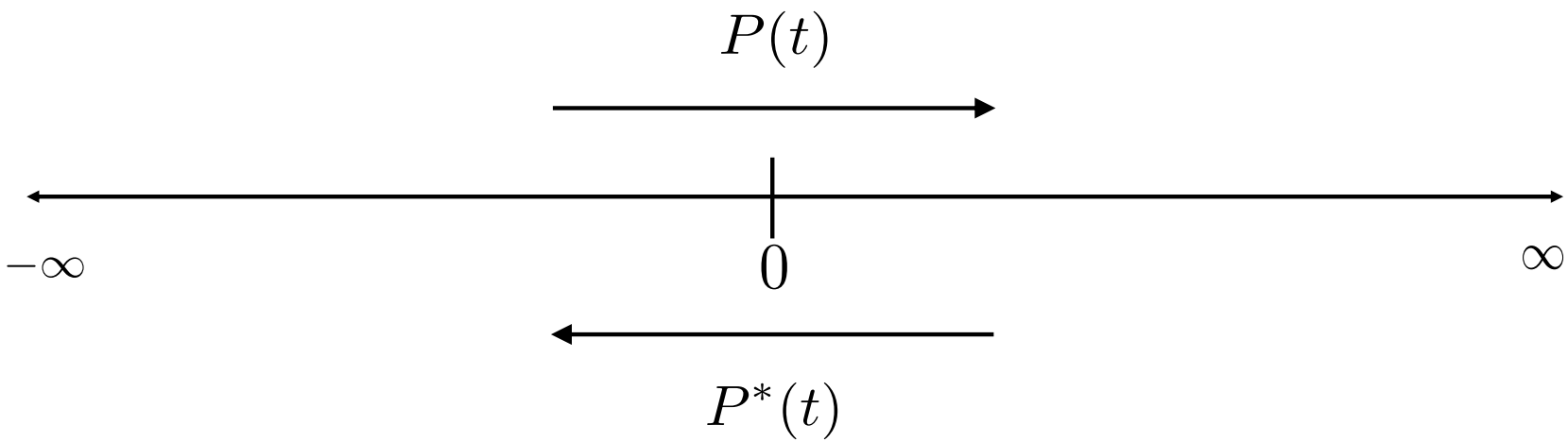
$G/G/1$ queue

- $T \equiv$ random variable of inter-arrival time with $E[T] = 1/\lambda$.
- $S \equiv$ random variable of service time with $E[S] = 1/\mu$.
- Results: no closed form — approximations only. See the handout *The $G/G/1$ and $G/G/m$ queueing systems* on PLMS (summarised in Companion Appendix A).

▷ Companion Appendix A: the $G/G/1$ and $G/G/m$ approximations of the PLMS handout (squared coefficients of variation C_a^2, C_s^2).

▶ Colab notebook: CP12 §8 ($G/G/1$ approximations) `[INCLASS] [full]`

Reversed process



▷ Companion Proof 6(a): the reversed chain has rates $q^*_{ij} = p_j q_{ji}/p_i$ — the derivation frame follows.

Reversed process

Transition probabilities of the process run backwards (stationary CTMC)

Let X be stationary, $P(X(t)=i)=p_i$ for all t. Looking *backwards* from time t:

$$P^*_{ij}(t) = P\big(X(0)=j \mid X(t)=i\big) = \frac{P(X(0)=j,\, X(t)=i)}{P(X(t)=i)}$$
$$= \frac{P(X(t)=i \mid X(0)=j)\,P(X(0)=j)}{P(X(t)=i)} = \frac{P_{ji}(t)\,p_j}{p_i}.$$

Hence $p_i\,P^*_{ij}(t) = p_j\,P_{ji}(t)$. Divide by t, let $t \downarrow 0$ ($P_{ij}(t)/t \to q_{ij}$ for $i \neq j$):

$$\lim_{t\downarrow 0} \frac{p_i\,P^*_{ij}(t)}{t} = \lim_{t\downarrow 0} \frac{p_j\,P_{ji}(t)}{t} \quad\Longrightarrow\quad p_i\,q^*_{ij} = p_j\,q_{ji}\,.$$

▷ The reversed process is a CTMC with rates $q^*_{ij} = p_j q_{ji}/p_i$; *time-reversible* means $q^*_{ij} = q_{ij}$.

Reversible process

Definition

A CTMC is said to be *time-reversible* if

$$p_i q_{ij} = p_j q_{ji}, \text{ i.e., } q^*_{ij} = q_{ij}.$$

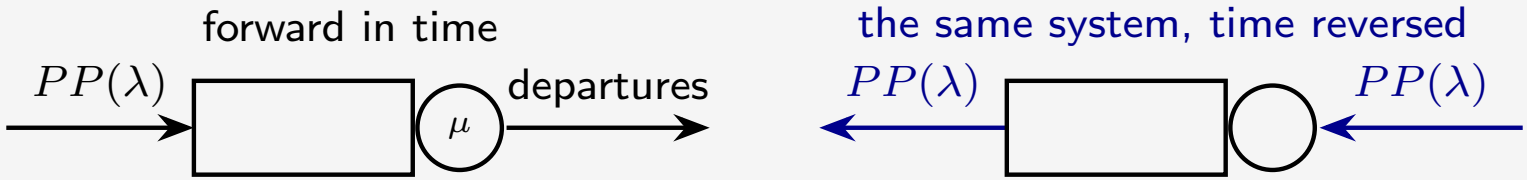
▷ Companion Proof 6(b): birth–death chains are reversible (cut equations = detailed balance); Burke's theorem.

Reversible process

Birth–death chains are time-reversible

$p_i q_{ij} = p_j q_{ji}$ are the **arc (detailed) balance** equations. In a birth–death chain the only arcs are $n \leftrightarrow n+1$, and the cut equations $\lambda_n p_n = \mu_{n+1} p_{n+1}$ *are* these equations: in steady state $M/M/1$, $M/M/s$, $M/M/\infty$ and their finite-capacity versions are all time-reversible.

Burke's theorem



Run the stationary $M/M/s$ backwards: its departures become the arrivals of the reversed process — which is the same process, so they form a $PP(\lambda)$: the departure process of a stationary $M/M/s$ is Poisson(λ), and the departures before t are independent of $X(t)$.

Truncation

- A: Time-reversible CTMC on $\mathcal{S}$
- B: Time-reversible CTMC on $\mathcal{S}'$
- $\mathcal{S}' \subset \mathcal{S}$, truncation of $\mathcal{S}$
- p^A: steady-state distribution of A
- p^B: steady-state distribution of B
- We can obtain p^B from p^A using the following:

$$p_j^B = \frac{p_j^A}{\sum_{i \in \mathcal{S}'} p_i^A}, \text{ for } j \in \mathcal{S}'.$$

▷ Companion Proof 6(c) (final-style): proof of the truncation theorem; $M/M/2/K$ from $M/M/2$, $M/M/K/K$ from $M/M/\infty$.

▶ Colab notebook: CP12 §4 ($M/M/2/K$ by truncation) `[INCLASS] [full]`

Truncation

Why truncation works (proof)

On $\mathcal{S}'$ keep every rate q_{ij} with $i, j \in \mathcal{S}'$ and delete the arcs that leave $\mathcal{S}'$. Candidate: $p_j^B = p_j^A / \sum_{i \in \mathcal{S}'} p_i^A$.

For $i, j \in \mathcal{S}'$ reversibility of A gives $p_i^A q_{ij} = p_j^A q_{ji}$; dividing by the constant $\sum_{\mathcal{S}'} p_i^A$:

$$p_i^B \, q_{ij} = p_j^B \, q_{ji} \qquad \text{— the detailed balance equations of } B.$$

Summing over i: $\sum_i p_i^B q_{ij} = p_j^B \sum_i q_{ji}$, i.e. the global balance $p^B Q_B = 0$. So p^B is the stationary distribution of B, and B is reversible too. □

▷ Nothing is solved: the weights of A are renormalised on $\mathcal{S}'$. Reversibility of A is essential — for a non-reversible chain the truncated weights are wrong.

Truncation

Final-style: $M/M/2/K$ from $M/M/2$ — without solving $pQ = 0$

$M/M/2$ weights (slide 14): $p_0 \propto 1$, $p_j \propto \frac{\lambda}{\mu}\left(\frac{\lambda}{2\mu}\right)^{j-1}$ for $j \geq 1$. Truncate to $\mathcal{S}' = \{0, 1, \ldots, K\}$:

$$p_j^{(K)} = \frac{\frac{\lambda}{\mu}\left(\frac{\lambda}{2\mu}\right)^{j-1}}{1 + \frac{\lambda}{\mu}\sum_{i=1}^{K}\left(\frac{\lambda}{2\mu}\right)^{i-1}} \quad (1 \leq j \leq K), \qquad p_0^{(K)} = \frac{1}{1 + \frac{\lambda}{\mu} \cdot \frac{1-(\lambda/2\mu)^K}{1-\lambda/2\mu}}$$

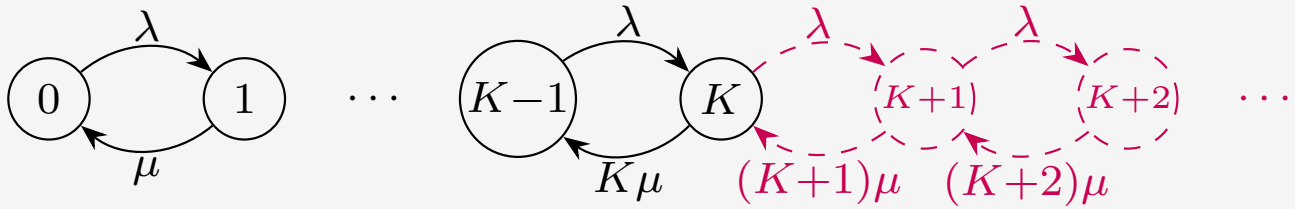
($\lambda \neq 2\mu$). Only the *ratios* of the $M/M/2$ weights enter, so the stability condition $\lambda < 2\mu$ is not needed.

$\lambda = \mu$, $K = 4$: $p^{(4)} = (0.348, 0.348, 0.174, 0.087, 0.043)$, loss $p_K = 4.3\,\%$.

▷ A 2025 final-exam question. CP12 §4 checks the answer against the direct solution of $pQ = 0$.

$M/M/\infty$ and $M/M/K/K$ queues

$M/M/K/K$ is the truncation of $M/M/\infty$ to $\mathcal{S}' = \{0, 1, \ldots, K\}$



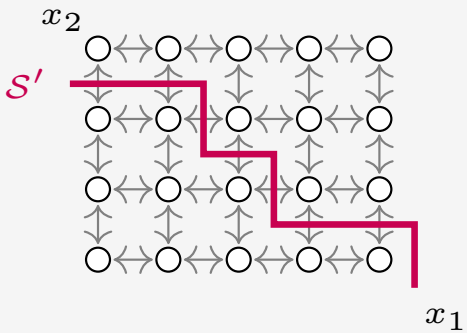
$M/M/\infty$ on $\mathcal{S} = \{0, 1, 2, \ldots\}$: $p_j^\infty = \dfrac{(\lambda/\mu)^j e^{-\lambda/\mu}}{j!}$, reversible (birth–death). Deleting the dashed part leaves exactly the $M/M/K/K$ diagram, so

$$p_j^K = \frac{p_j^\infty}{\sum_{i=0}^K p_i^\infty} = \frac{(\lambda/\mu)^j e^{-\lambda/\mu}/j!}{\sum_{i=0}^K (\lambda/\mu)^i e^{-\lambda/\mu}/i!} = \frac{(\lambda/\mu)^j/j!}{\sum_{i=0}^K (\lambda/\mu)^i/i!}$$

— Erlang's loss distribution in one line.

$M/M/\infty$ and $M/M/K/K$ queues

Truncation is not only for intervals

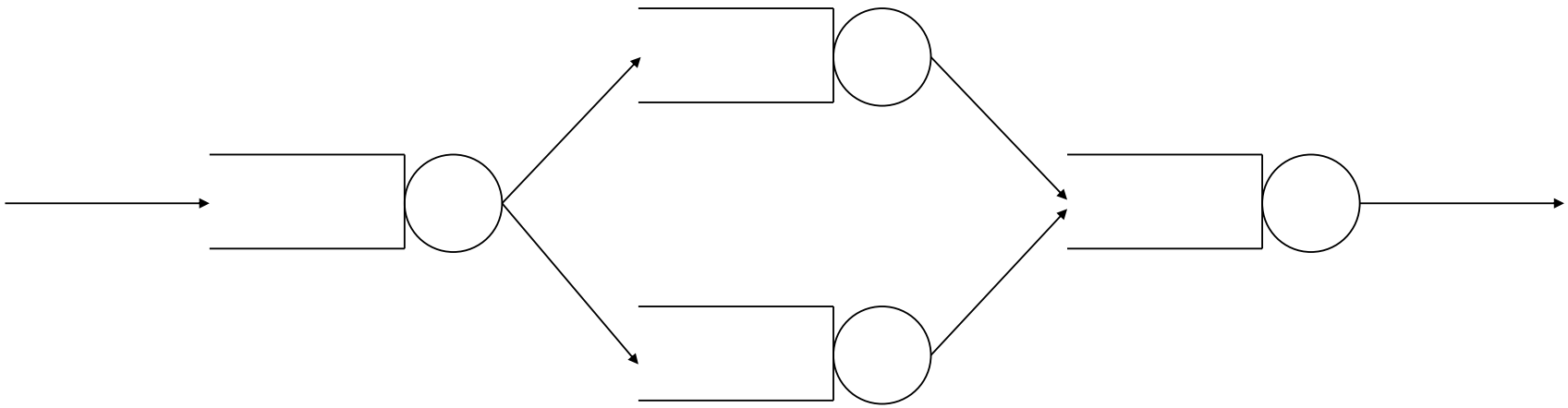


Any subset $\mathcal{S}'$ of the state space of a reversible chain works. Example: two *independent* $M/M/\infty$ queues, state (x_1, x_2) with product weights $p_1(x_1)p_2(x_2)$ (a product of reversible chains is reversible), restricted to $x_1 + x_2 \le K$ (two call types sharing K lines) or to the staircase on the left:

$$p^B(x_1, x_2) = \frac{p_1(x_1)\, p_2(x_2)}{\sum_{\mathcal{S}'} p_1(y_1)\, p_2(y_2)}, \quad (x_1, x_2) \in \mathcal{S}'.$$

▷ The workhorse behind loss networks and many blocking models — without a single balance equation.

Network of queues



- Network types
  - Open: external arrivals, departures
  - Closed: conservation of customers

▷ Companion §1.6: the state of a network is the vector $X(t) = (X_1(t), \ldots, X_N(t))$.

Jackson network

- Single-class open queueing network
- External arrivals are Poisson.
- Service times are exponential.
- Any number of servers in a station, including ∞.

▷ Companion Proof 8: traffic equations, product form, tandem and feedback queues, Braess' paradox.

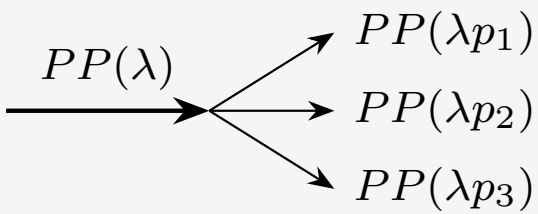
Jackson network

Two more standing assumptions

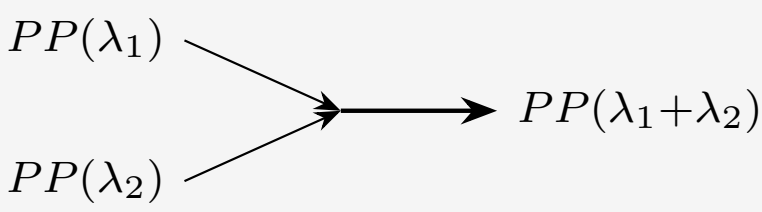
Capacity of each station is infinite (nobody is blocked or lost inside the network); all stations are stable. Routing is Markovian: after service at i go to j w.p. p_{ij} or leave w.p. $r_i = 1 - \sum_j p_{ij}$, independently of everything else.

The three tools that make the analysis work

1) Poisson splitting

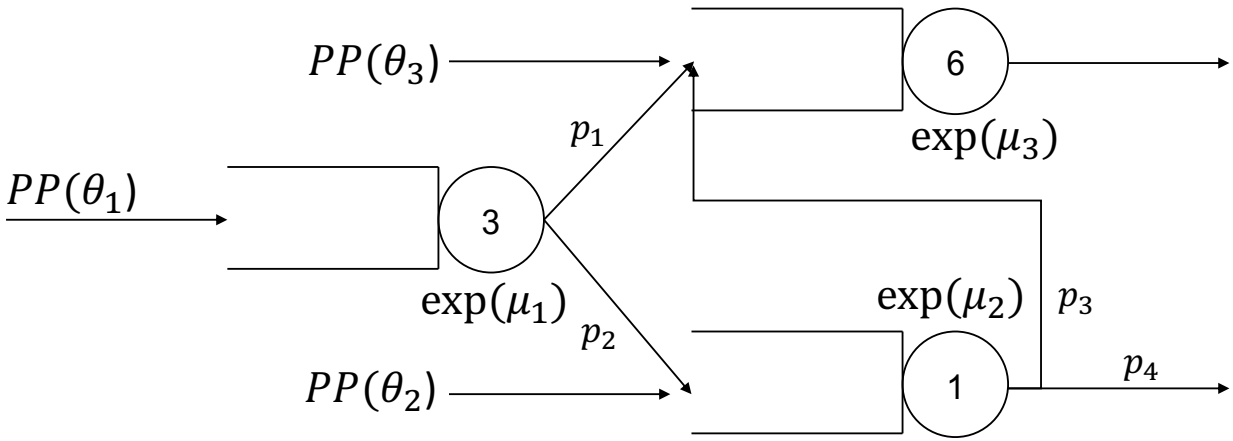


2) Merging



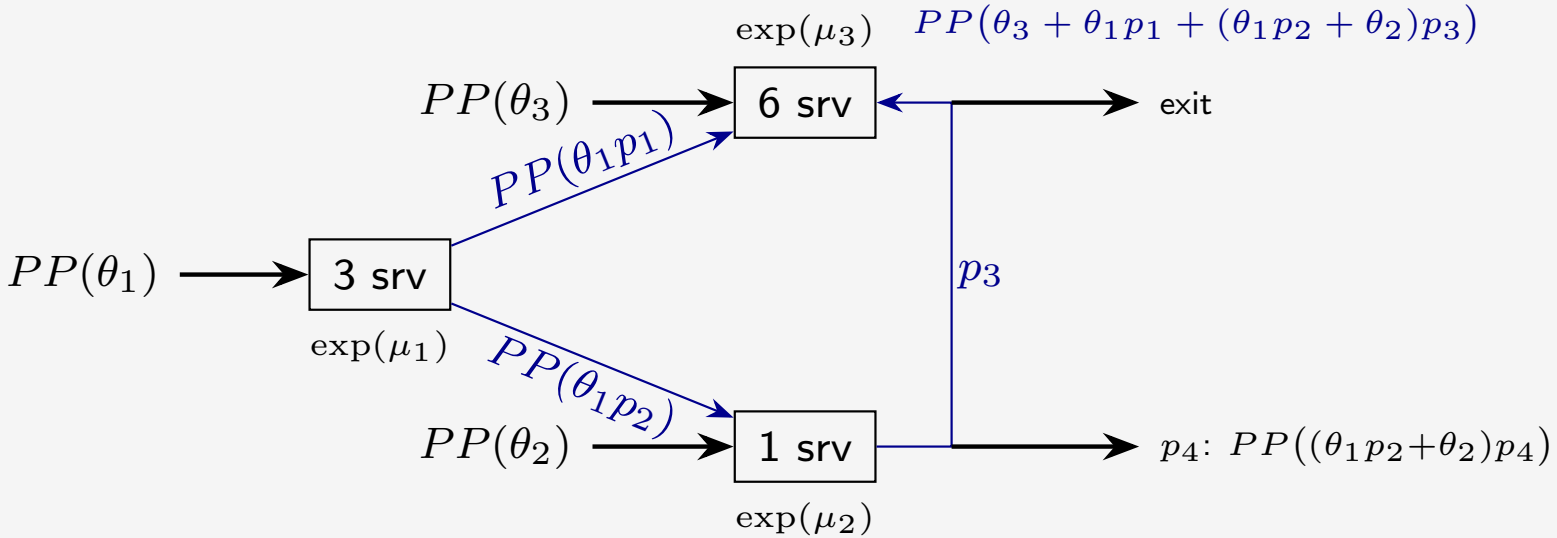
3) Time reversibility (Burke): the departure process of each $M/M/s$ station is PP in steady state — every station receives a superposition of thinned Poisson flows. (Caveat: with feedback the internal flows are not Poisson, yet the product form holds — CP13 §4.)

Example


$PP(\theta_3)$
$PP(\theta_1)$
$PP(\theta_2)$
3
6
1
$\exp(\mu_1)$
$\exp(\mu_3)$
$\exp(\mu_2)$
p_1
p_2
p_3
p_4

Example

Effective arrival rates: follow the Poisson flows through the picture



Node 2 receives a split flow merged with an external flow, $PP(\theta_1 p_2 + \theta_2)$; its departures are Poisson (Burke) and split again.

Effective arrival rates:

$$\lambda_1 = \theta_1, \qquad \lambda_2 = \theta_1 p_2 + \theta_2, \qquad \lambda_3 = \theta_3 + \theta_1 p_1 + (\theta_1 p_2 + \theta_2)\, p_3 \,.$$

Wanted: $\lim_t P(X_1(t) = i, X_2(t) = j, X_3(t) = k)$ — a CTMC on $\mathbb{Z}_+^3$ that is never solved directly.

Setting

- N nodes (stations)
- Each node i has $PP(\theta_i)$ as external arrivals. If no external arrival, $\theta_i = 0$.
- There are s_i servers at node i, $s_i \in \{1, 2, \ldots\} \cup \{\infty\}$.
- Each node has infinite capacity.
- Service time at node i is $\exp(\mu_i)$.
- When a service is completed at node i, the customer joins node j with prob. p_{ij} and exit with prob. r_i.

$$\sum_{j=1}^{N} p_{ij} + r_i = 1.$$

▷ Companion Proof 8(a): the routing matrix $P = [p_{ij}]$ is substochastic — $I - P$ is invertible when every customer eventually leaves.

Analysis

- Assumptions
  - All stations are stable.
  - At least one node has external arrivals.
  - At least one node has departures.
- For node i,

$$\lambda_i = \theta_i + \sum_{j=1}^{N} \lambda_j p_{ji}.$$

  λ_i is called *effective arrival rate* to node i.
- Matrix form

$$\boldsymbol{\lambda} = \boldsymbol{\theta} + \boldsymbol{\lambda} P,$$

  where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)$, $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)$, and $P = [p_{ij}]$.

$$\boldsymbol{\lambda} = \boldsymbol{\theta}(I - P)^{-1}.$$

Analysis

- Node i can be modeled as an $M/M/s_i$ queue with $PP(\lambda_i)$ arrivals and exp(μ_i) service.
- In steady state, the number of customers at node i, $X_i(t)$, is independent of the number of customers at any other nodes.

$$\lim_{t\to\infty} P\left(X_1(t) = x_1, \ldots, X_N(t) = x_N\right)$$
$$= \lim_{t\to\infty} P\left(X_1(t) = x_1\right) \cdots P\left(X_N(t) = x_N\right).$$

▷ Companion Proof 8(b): the product form, proved by substituting the guess $p(\mathbf{x}) = C\prod_i \phi_i(x_i)$ into the global balance equations (the Independence handout).

▶ Colab notebook: CP13 §5 (split vs. pooled servers) `[INCLASS] [full]`

Analysis

Using the product form: node by node, then the whole network

Traffic equations $\boldsymbol{\lambda}(I - P) = \boldsymbol{\theta}$ first ($I - P$ is invertible because every customer eventually leaves). Then **node** i = an $M/M/s_i$ queue with arrival rate λ_i (the effective rate, *not* θ_i): $p_i(x)$, L_i from slides 17–22;

$$L_i = \frac{\lambda_i}{\mu_i - \lambda_i} \ (s_i = 1), \qquad L_i = \frac{\lambda_i}{\mu_i} \ (s_i = \infty), \qquad W_i = \frac{L_i}{\lambda_i}$$

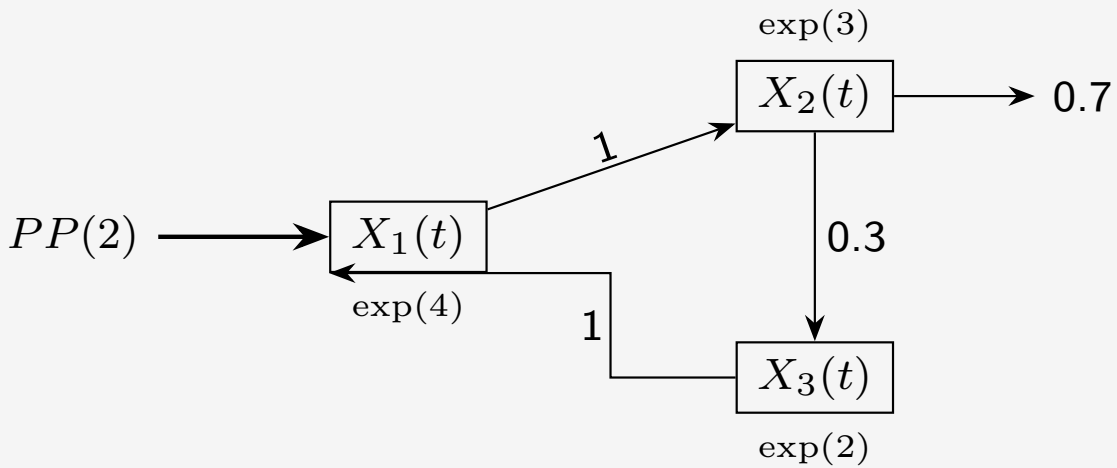
(Little at node i: W_i is the time per *visit*). **Whole network** (Little for a box around everything, arrival rate $\sum_i \theta_i$):

$$L = \sum_{i=1}^{N} L_i, \qquad W = \frac{L}{\sum_{i=1}^{N} \theta_i} \qquad \Big(= \sum_i v_i W_i, \quad v_i = \frac{\lambda_i}{\sum_k \theta_k} \Big).$$

▷ The X_i are independent in steady state even where the flow into node i is not Poisson; only the stationary law factorises, not the process (Companion Proof 8).

Example

A three-station network with feedback: $PP(2)$ into station 1, service rates $4, 3, 2$



$$\boldsymbol{\theta} = (2, 0, 0), \quad P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0.3 \\ 1 & 0 & 0 \end{pmatrix}$$

Traffic equations: $\lambda_1 = 2 + \lambda_3, \;\; \lambda_2 = \lambda_1, \;\; \lambda_3 = 0.3\lambda_2 \Rightarrow \lambda_1 = 2/0.7$:

$$\boldsymbol{\lambda} = \boldsymbol{\theta}(I - P)^{-1} = (2.8571,\ 2.8571,\ 0.8571)\,.$$

▷ All three $\rho_i = \lambda_i/\mu_i$ $(0.714,\ 0.952,\ 0.429)$ are below 1: stable, with station 2 as the bottleneck.

▶ Colab notebook: CP13 §1 (three-station network) `[INCLASS] [full]`

Example

Each node is $M/M/1$ with its own λ_i

$$L_1 = \frac{2.8571}{4 - 2.8571} = 2.5, \qquad L_2 = \frac{2.8571}{3 - 2.8571} = 20,$$

$$L_3 = \frac{0.8571}{2 - 0.8571} = 0.75; \qquad W_i = \frac{L_i}{\lambda_i}: \quad W_1 = 0.875, \quad W_2 = 7, \quad W_3 = 0.875$$

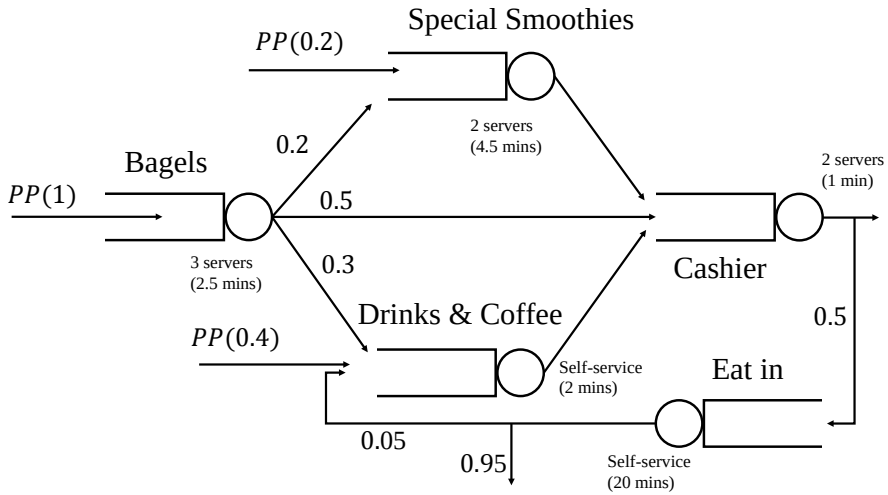
(per visit).

The network

$L = L_1 + L_2 + L_3 = 23.25$ customers; $W = L/\theta_1 = 23.25/2 = 11.63$ time units per customer.

Check via visits: a customer visits stations 1 and 2 on average $1/0.7 = 1.43$ times and station 3 0.43 times, and $1.43\,(0.875 + 7) + 0.43 \cdot 0.875 = 11.63$ ✓.

Example: Bay-Gull Bagels (handout)



Bay-Gull Bagels is a bagel store in downtown Pohang. All arrival processes are Poisson and all service times exponential (rates per minute, service times in minutes); a "self-service" station has infinitely many servers; all queues have infinite capacity. Model the store as a Jackson network and obtain the average number of customers at each of the 5 stations and in the whole store.

▶ Colab notebook: CP13 §2 (Bagels: product form vs. simulation) `[INCLASS]` `[full]`

Example: Bay-Gull Bagels (handout)

Step 1: traffic equations

Stations: 1 Bagels ($s = 3$, $1/\mu = 2.5$), 2 Smoothies ($s = 2$, 4.5), 3 Drinks ($s = \infty$, 2), 4 Cashier ($s = 2$, 1), 5 Eat in ($s = \infty$, 20);
$\boldsymbol{\theta} = (1,\ 0.2,\ 0.4,\ 0,\ 0)$ per minute.

$$\lambda_1 = 1, \qquad \lambda_2 = 0.2 + 0.2\lambda_1, \quad \lambda_3 = 0.4 + 0.3\lambda_1 + 0.05\lambda_5,$$
$$\lambda_4 = 0.5\lambda_1 + \lambda_2 + \lambda_3, \quad \lambda_5 = 0.5\lambda_4.$$

Substituting, $\lambda_4 = 0.9 + \lambda_3$, $\lambda_5 = 0.45 + 0.5\lambda_3$, $\lambda_3 = 0.7225 + 0.025\lambda_3$, so

$$\boldsymbol{\lambda} = (1,\ 0.4,\ 0.741,\ 1.641,\ 0.821) \text{ per minute}.$$

Step 2: stability

$$\rho_1 = \frac{1}{3 \cdot 0.4} = 0.833, \qquad \rho_2 = \frac{0.4 \cdot 4.5}{2} = 0.9, \qquad \rho_4 = \frac{1.641}{2} = 0.821 \text{ —}$$

all below 1; the self-service stations are always stable.

Example: Bay-Gull Bagels (handout)

Step 3: five independent $M/M/s$ queues

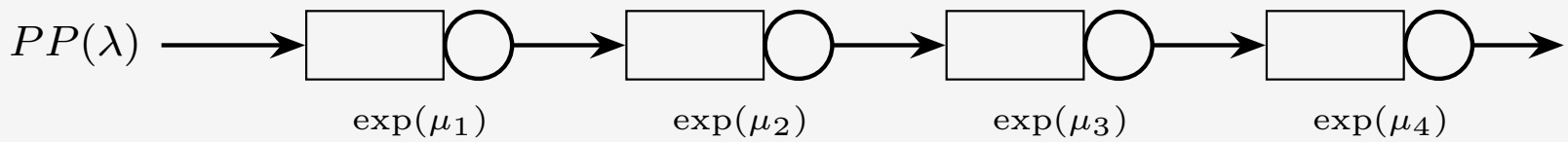
station	model	$a_i = \lambda_i/\mu_i$	ρ_i	L_i
1 Bagels	$M/M/3$	2.5	0.833	6.01
2 Smoothies	$M/M/2$	1.8	0.9	9.47
3 Drinks & Coffee	$M/M/\infty$	1.482	–	1.48
4 Cashier	$M/M/2$	1.641	0.821	5.02
5 Eat in	$M/M/\infty$	16.41	–	16.41
whole store				$L = 38.4$

$M/M/\infty$: $L_i = a_i$; $M/M/s$: $L_i = L_{q,i} + a_i$ with the L_q of slide 19. $W = L/(1 + 0.2 + 0.4) = 24.0$ min per customer. Smoothies ($\rho = 0.9$) holds more people than Bagels although it carries less than half the flow.

▷ CP13 §2 confirms the table by a discrete-event simulation of the store — and shows what breaks when a service time is made deterministic.

Tandem queue

Stations in series: the simplest Jackson network



$p_{i,i+1} = 1$, $r_N = 1$, $\boldsymbol{\theta} = (\lambda, 0, \ldots, 0)$: the traffic equations give $\lambda_i = \lambda$ at every station — what goes in must come out. By Burke each station sees Poisson(λ) input, and the product form gives

$$P(X_1 = x_1, \ldots, X_N = x_N) = \prod_{i=1}^{N} (1 - \rho_i)\, \rho_i^{x_i}, \qquad \rho_i = \frac{\lambda}{\mu_i} < 1.$$

▷ The numbers at the stations are independent at a fixed time although every customer passes through all of them — the surprise of Burke's theorem.

▶ Colab notebook: CP13 §3 (tandem: Burke's theorem) `[INCLASS] [full]`

Tandem queue

L and W

$$L = \sum_{i=1}^{N} \frac{\lambda}{\mu_i - \lambda}, \qquad W = \frac{L}{\lambda} = \sum_{i=1}^{N} \frac{1}{\mu_i - \lambda}$$

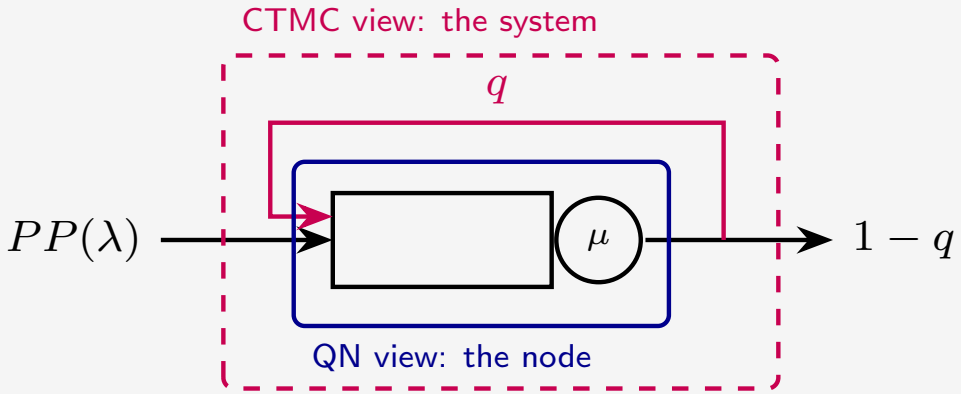
— the sojourn is the sum of the N station sojourns, each $\exp(\mu_i - \lambda)$ as in an isolated $M/M/1$ (for single servers they are even independent: Reich's theorem).

Example

$\lambda = 2$, $\mu = (4, 3)$: $L = \frac{2}{4-2} + \frac{2}{3-2} = 1 + 2 = 3$, $W = 1.5$ — the slower second station carries two thirds of the congestion.

Feedback queue

One station, two viewpoints



After each service the customer leaves w.p. $1-q$ or rejoins the queue w.p. q.

1) CTMC. The number in the system is a birth–death chain with birth rate λ and death rate $\mu(1-q)$ (a fed-back service does not change the state): an $M/M/1$ with service rate $\mu(1-q)$,

$$L = \frac{\lambda}{\mu(1-q)-\lambda}, \qquad W = \frac{1}{\mu(1-q)-\lambda}.$$

▶ Colab notebook: CP13 §4 (feedback queue) `[INCLASS] [full]`

Feedback queue

2) Queueing network view

One node with $p_{11} = q$: $\hat{\lambda} = \lambda + q\hat{\lambda}$, so $\hat{\lambda} = \lambda/(1-q)$ (external + fed-back customers); the node is $M/M/1$ with rates $\hat{\lambda}$, μ:

$$L = \frac{\hat{\lambda}}{\mu - \hat{\lambda}} = \frac{\lambda/(1-q)}{\mu - \lambda/(1-q)} = \frac{\lambda}{\mu(1-q) - \lambda} \checkmark,$$

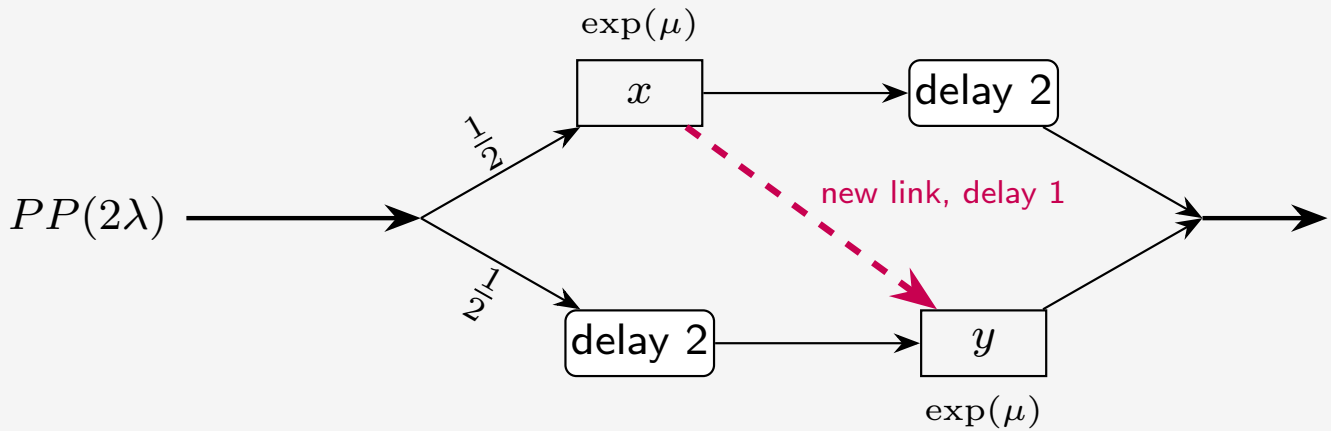
$$W_{\text{node}} = \frac{L}{\hat{\lambda}} = \frac{\lambda}{\mu(1-q) - \lambda} \cdot \frac{1-q}{\lambda} = \frac{1-q}{\mu(1-q) - \lambda}.$$

Same L, two different W's

The QN W is per *visit* (one pass through the node), the CTMC W is per *customer*: a customer makes $1/(1-q)$ visits on average, and $W_{\text{customer}} = W_{\text{node}}/(1-q)$. Always ask which arrival rate Little's law is applied with. (The flow into the node, external + feedback, is not Poisson — the product form does not need it to be.)

Braess' paradox

Two parallel routes, each a queue plus a fixed delay of 2



Before the link. Half of the traffic takes each route: each queue is $M/M/1$ with rate λ (Poisson splitting), sojourn $1/(\mu-\lambda)$,

$$W_x = W_y = \frac{1}{\mu-\lambda} + 2 \quad \Longrightarrow \quad W = \frac{1}{\mu-\lambda} + 2\,.$$

▷ x and y also denote the mean sojourn times at the two queues; each depends on the flow it receives, $x = 1/(\mu - f_x)$.

▶ Colab notebook: CP13 §6 (Braess' paradox) `[INCLASS] [full]`

Braess' paradox

After the link: customers choose the fastest route for themselves

Three routes: $x \to 2$, $x \to 1 \to y$, $2 \to y$. In equilibrium every used route takes the same time (nobody can gain by switching):

$$x + 2 \;=\; x + 1 + y \;=\; 2 + y \quad\Longrightarrow\quad x = y = 1 \quad\Longrightarrow\quad W = 3\,.$$

($x = 1$ means $1/(\mu - f_x) = 1$: each queue carries flow $\mu - 1$; all three routes are used when $\lambda + 1 < \mu \le 2\lambda + 1$.)

The paradox

If $\mu > \lambda + 1$: $W_{\text{before}} = \dfrac{1}{\mu - \lambda} + 2 \;<\; 3 = W_{\text{after}}$ — adding a free, fast link made *everyone* slower. $\lambda = 1$, $\mu = 2.5$: $2.67 \to 3$.

▷ Selfish routing is not socially optimal; a central planner would ignore the new link. CP13 §6 computes the flows and simulates both networks.